STACKS, DISCRIMINANTS: LEMMA-SYNTOMIC-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2281-2299 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 2301-2323, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 24 of queue_discriminant.json (pb=1098, stmt 2281-2299, proof 2301-2323).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07CF: sources/stacks-discriminant/pinned-original-source.tex, source lines 2323–2323. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02M9: sources/stacks-discriminant/pinned-original-source.tex, source lines 2323–2323. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BWE: sources/stacks-discriminant/pinned-original-source.tex, source lines 2005–2026. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : Y \to X$ be a morphism of schemes. The following are equivalent*

(1) *f is finite and syntomic,*
(2) *f is finite, flat, and a local complete intersection morphism,*
(3) *f is finite, flat, locally of finite presentation, and the fibres of f are local complete intersections,*
(4) *f is finite and for every $x \in X$ there is an affine open $x \in U = \mathrm{Spec}(A) \subset X$ an integer n and $f_1, \ldots, f_n \in A[x_1, \ldots, x_n]$ such that $f^{-1}(U)$ is isomorphic to the spectrum of $A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$,*
(5) *f is finite, flat, locally of finite presentation, and $NL_{X/Y}$ has tor-amplitude in $[-1, 0]$, and*

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(6) *f is finite, flat, locally of finite presentation, and $NL_{X/Y}$ is perfect of rank 0 with tor-amplitude in $[-1, 0]$,*

Proof. The equivalence of (1), (2), (3), (5), and (6) and the implication (4) $\Rightarrow$ (1) follow immediately from Lemma 0BWE. Assume the equivalent conditions (1), (2), (3), (5), (6) hold. Choose a point $x \in X$ and an affine open $U = \mathrm{Spec}(A)$ of x in X and say x corresponds to the prime ideal $\mathfrak{p} \subset A$. Write $f^{-1}(U) = \mathrm{Spec}(B)$. Write $B = A[x_1, \ldots, x_n]/I$. Since $NL_{B/A}$ is perfect of tor-amplitude in $[-1, 0]$ by (6) we see that I/I^2 is a finite locally free B-module of rank n. Since $B_{\mathfrak{p}}$ is semi-local we see that $(I/I^2)_{\mathfrak{p}}$ is free of rank n, see Algebra, Lemma 02M9. Thus after replacing A by a principal localization at an element not in $\mathfrak{p}$ we may assume I/I^2 is a free B-module of rank n. Thus by Algebra, Lemma 07CF we can find a presentation of B over A with the same number of variables as equations. In other words, we may assume $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_n)$. This proves (4). $\square$